

Article

Fluoro-Silicon-Modified Polythiourethane Copolymer for Marine Antifouling Coatings

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Abstract: Traditional marine antifouling coatings function through releasing toxic antifouling agents, causing serious harm to marine ecosystems. To address this challenge, an eco-friendly fluoro-silicon-modified polythiourethane (FSi-PTU) coating has been prepared via a polymerization reaction with dihydroxy propyl silicone oil (HO-PDMS-OH), 1H,1H,2H,2H-perfluorohexanol (FTOH), hexamethylene diisocyanate (HDI), and pentaerythritol tetrakis (3-mercaptopropionate) (PETMP). The FSi-PTU polymer incorporates siloxane segments and fluorinated side chains, which are inhomogeneously distributed on the coating surface and construct a hydrophobic surface. The FSi-PTU coating exhibits good hydrophobicity, strong adhesion (≥ 2.14 MPa), and improved mechanical properties. The antifouling properties of the FSi-PTU coating have been researched. The results of laboratory tests demonstrate that the FSi-PTU coating exhibits excellent anti-protein adsorption and anti-algal attachment performance. The FSi-PTU-2 coating shows certain antifouling properties in the actual seawater test for three months. The results provide a certain reference value for developing eco-friendly marine antifouling coatings.

Keywords: fluoro-silicon; polythiourethane; low surface energy; antifouling coatings



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1. Introduction

Marine biofouling is one of the primary challenges to the development of the marine industry [1–3]. Biofouling causes severe negative impacts on marine vessels. It speeds up hull corrosion, curtails ship lifetimes, raises fuel consumption and maintenance costs, and even causes biological invasion [4,5]. Marine antifouling coatings are regarded as the most effective, economical, and widely used antifouling strategy [6,7]. Unfortunately, most antifouling coatings release biocides and toxic substances, which cause serious harm to marine ecosystems. The International Convention on the Control of Harmful Antifouling Systems on Ships (AFS) ordered that the use of TBT as fungicides in ship antifouling systems be globally prohibited before 1 January 2003, and that a total ban be implemented before 1 January 2008 [8,9]. At present, self-polishing antifouling coatings contain 40%–50% copper oxide and a small amount of organic biocides (pyrithione (CuPT), zinc pyrithione (ZnPT), and isothiazolinone) [10]. Studies have revealed that the embryo and larval phases of mussels, oysters, and sea urchins exhibit high sensitivity to copper exposure. Specifically, the EC50 values for total dissolved copper in these organisms are 6.8 µg/L for mussels,

12.1 $\mu\text{g/L}$ for oysters, and 14.3 $\mu\text{g/L}$ for sea urchins. The EC/LC50 concentrations for fish, when exposed to total dissolved copper, fall within the range of 0.12 to 1.5 $\mu\text{g/L}$ [10–12]. Rainbow trout larval development and growth are affected by CuPT (from 0.5 $\mu\text{g/L}$) [13]. Copper and organic biocides also have a detrimental impact on the marine environment, and their use in marine antifouling coatings may be restricted in the future [14,15]. The EU Biocidal Products Regulation (BPR, (EU) 2021/1099) stipulates that the copper ion release rate from antifouling coatings in dynamic seawater must not exceed 4 $\mu\text{g/cm}^2/\text{day}$; this stipulation will be comprehensively enforced on 1 January 2026. Non-toxic fouling-release coatings have become a major research direction in the field of marine antifouling coatings [16]. There is an urgent demand to develop eco-friendly antifouling coatings to replace traditional, toxic antifouling coatings.

Fouling release coatings are ideal eco-friendly antifouling materials, given they have low surface energy (SE) and elastic modulus. These properties enable the coatings to remove biofouling organisms from their surface by shear force without the need to release any toxic antifouling agents [2,17,18]. However, the practical application of traditional silicone elastic coatings is restricted because of their poor adhesion strength, weak mechanical performance, and poor static antifouling ability [19,20]. The primary aim of the current study was to improve the adhesion strength and mechanical performance of the coating. Typically, the drawbacks of poor adhesion and weak mechanical properties of the coatings can be addressed by introducing polyurethane (PU), acrylic resin, and epoxy resin [7,21]. For example, Wang [22] et al. synthesized silicone-modified polyaspartic ester polyurea with a low SE and elasticity modulus via simultaneous cross-linking polymerization. The synergy between silicone and polyurea provides excellent mechanical properties and fouling release properties. Zhang [23] et al. synthesized a silicone low SE coating with zwitterionic side chains via a polymerization reaction. The coating had good mechanical properties, higher than 6.0 MPa, retaining the original fouling-release characteristics of the silicone materials and effectively preventing the adhesion of *Pseudomonas* sp. and the accumulation of biofilms and diatoms. Lin [24] et al. synthesized a silicone-based polybenzoxazine coating with a low modulus and low SE using natural phenolic derivatives and poly-(dimethylsiloxane)etherimide, which contains phenolic hydroxyl and hydrophobic siloxane groups. The coating shows a strong bond to the substrates; moreover, it demonstrates good fouling-release performance against bacteria and algae. Therefore, adding polymer resins to the silicone elastomer effectively enhances the mechanical properties and adhesion strength of the polymer coatings.

Polydimethylsiloxane (PDMS) effectively promotes the removal of biofouling organisms, but it has a poor static antifouling performance in seawater [25–27]. Fluorinated polymers are widely used for designing marine antifouling coatings due to their low SE, excellent hydrophobicity, and environmental friendliness [28,29]. Due to the low SE and excellent hydrophobicity of fluorinated polymers, it is difficult for microorganisms to spread secretions on the coating surface, which inhibits the effective adhesion of marine fouling organisms [30,31]. In this respect, fluorinated polymers, as another class of low SE coating materials, showed better antifouling properties compared with PDMS coating. Over recent years, some research has been conducted to integrate PDMS with fluorinated groups to enhance its antifouling properties. For example, Xu [32] et al. synthesized a poly(dimethyl-methylphenyl-methyltrifluoropropyl)siloxane coating with a low SE and Young's modulus by the hydrolysis–polycondensation method with dimethyl dimethoxy silane, phenyl methyl dimethoxy silane, and 3,3,3-trifluoropropyl methyl dimethoxy silane. The coating demonstrates good thermal stability, mechanical properties, and antifouling properties. Xiang [33] et al. prepared fluorinated block copolymer/polystyrene blend coatings through a two-step atom transfer radical polymerization reaction. The coatings

displayed good compatibility and durable antifouling properties. The fluorinated side chains of the polymer should organize into an ordered structure, and the density of $-CF_3$ groups on the coating surface should remain high, creating a surface with a low SE and improving the antifouling properties of the coating [28,30,34]. Therefore, it is necessary to develop a study on fluoro-silicon-modified polymer coatings.

In this study, an FSi-PTU coating was prepared with HO-PDMS-OH, FTOH, HDI, and PETMP via a polymerization reaction. HO-PDMS-OH and FTOH acted as the main and side chains of the FSi-PTU polymer, respectively, to reduce the surface energy of the coating and improve its antifouling performance. In addition, the FSi-PTU coatings were covalently connected to form cross-linked networks by using PETMP, improving the mechanical properties of the coatings. The structures of the FSi-PTU polymers, along with the elemental distribution and three-dimensional morphology of the coating surfaces, were characterized. The hydrophobicity, adhesion strength, mechanical properties, anti-protein adsorption properties, and anti-algal attachment properties of the FSi-PTU coatings were researched. The antifouling properties of the FSi-PTU coatings were investigated during an actual marine test that took three months in the South China Sea.

2. Materials and Methods

2.1. Materials

Dihydroxy propyl silicone oil ($M_n = 2000$) was purchased from Anhui Aiyota Silicone Oil Co., Ltd. (Bengbu, China). 1H,1H,2H,2H-perfluorohexanol (99%), Hexamethylene diisocyanate (99%), pentaerythritol tetrakis (3-mercaptopropionate) (99%), diiodomethane (98%), and ditin butyl dilaurate (DBTDL, 95%) were received from Aladdin Corporation (Shanghai, China). Bovine serum albumin-FITC (BSA-FITC, 98%) was obtained from Beijing Bersee Science and Technology Co., Ltd. (Beijing, China). Artificial seawater (ASW) was obtained from Jiangsu Maige Biotechnology Co., Ltd. (Taizhou, China). Phosphate-buffered solution (PBS, PH 7.4) was purchased from Beijing EallBio Biomedical Technology Co., Ltd. (Beijing, China). Ethyl acetate (99.5%) was purchased from ChengDu Chron Chemicals Co., Ltd. (Chengdu, China). Sylgard 184 was obtained from Dow Corning (Midland, MI, USA). Navicula and Chlorella vulgaris were purchased from Henan Zeju Biotechnology Co., Ltd. (Zhengzhou, China).

2.2. Preparation of Si-PTU Marine Antifouling Coatings

The Si-PTU was prepared with HO-PDMS-OH, HDI, and PETMP (Figure 1a). HO-PDMS-OH (0.1 g) and DBTDL (0.003 g) were dissolved in ethyl acetate (8 mL). The mixture was transferred into a conical flask. HDI (0.336 g) was added slowly to the conical flask under ultrasonic conditions after being subjected to a reaction for 30 min. Subsequently, PETMP (0.489 g) was dissolved in ethyl acetate (2 mL) and transferred into the conical flask. The reaction was continued under ultrasonic conditions for 30 min at 35 ± 2 °C. The temperature variation was continuously monitored using the built-in temperature detector of the ultrasonic cleaner. To maintain the water temperature at a stable temperature of 35 ± 2 °C, cold water was added slowly as needed.

2.3. Preparation of FSi-PTU-x Marine Antifouling Coatings

The FSi-PTU-x was prepared with HO-PDMS-OH, FTOH, PETMP, and HDI (Figure 1b). HO-PDMS-OH (0.1 g), FTOH (0.066, 0.099, 0.132, and 0.165 g), and DBTDL (0.003 g) were dissolved in ethyl acetate (8 mL), and the mixture was transferred into a conical flask. HDI (0.336 g) was added slowly to the conical flask under ultrasonic conditions, after a reaction had taken place for 30 min. Subsequently, PETMP (0.489 g) was dissolved in ethyl acetate (2 mL) and transferred into the conical flask. The reaction was continued under ultrasonic

conditions for 30 min at $35 \pm 2^\circ\text{C}$. The temperature variation was continuously monitored using the built-in temperature detector of the ultrasonic cleaner. To maintain the water temperature at a stable temperature of $35 \pm 2^\circ\text{C}$, cold water was added slowly as needed.

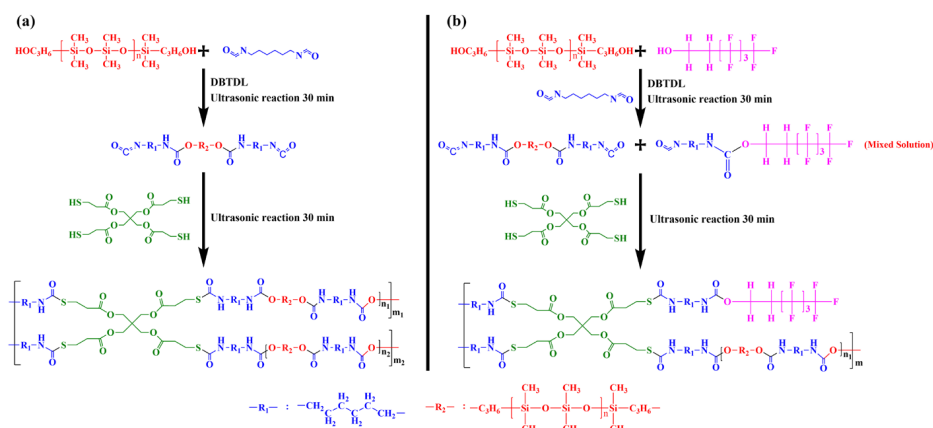


Figure 1. Synthesis of the Si-PTU (a) and FSi-PTU (b) coatings.

2.4. Characterization

FT-IR spectra were measured using a Fourier transform infrared spectrometer (Thermo Fisher Scientific, Nicolet iS5, Waltham, MA, USA). The elemental distribution of the coating surfaces was characterized by an energy-dispersive spectrometer (EDS, IXRF, Model 550i, Austin, TX, USA) used in conjunction with a scanning electron microscope (SEM, Hitachi, S-3400N, Tokyo, Japan). The 3D surface topography and surface roughness were analyzed using an atomic force microscope (AFM, Bruker, Dimension ICON, Billerica, MA, USA). The contact angle and the corresponding SE of the coatings were acquired via the use of a Drop Shape Analyzer (KRÜSS, DSA25S, Hamburg, Germany). Tensile tests were carried out using a universal testing machine (CMT4104, MTS Systems, Shanghai, China).

2.5. Adhesion Tests

The adhesion strength of the coatings was measured using a digital pull-off adhesion tester (BGD 500/S, Biuged Precision Instruments Co., Ltd., Guangzhou, China). Five aluminum dollies with a diameter of 20 mm were adhered to the surface of the epoxy resin plate using polymers as adhesives. Five different regions were tested for each polymer to obtain an average value. The pressurization rate was 0.1 MPa/s. All the tests were conducted by the same operator to minimize the standard error.

2.6. Anti-Protein Adsorption Experiment

All the coatings were immersed in a BSA-FITC/PBS (0.50 mg/L) solution and rinsed with PBS solution after 12 h. The fluorescence signals of the coatings were observed by using an upright fluorescence microscope (Nikon Eclipse 50i, Tokyo, Japan). Five different regions were tested for each coating. We used Image J (1.52a) software to measure the fluorescent area coverage of the corresponding fluorescent images and calculate the average of these fractions as the statistical result. All the experimental procedures were completed at room temperature.

2.7. Navicula and Chlorella Vulgaris Colonization Assays

Navicula (1×10^5 cells/mL) and Chlorella vulgaris (1×10^5 cells/mL) suspensions were prepared with ASW. All the coatings were immersed in algal suspension (15 mL) and incubated in an incubator (with a light–dark cycle, under the illumination of an 8 w light source, at $26 \pm 2^\circ\text{C}$) and then rinsed with deionized water after 14 days of incubation. An

optical microscope (Zeiss, Axiovert 5, Jena, Germany) was used to observe the adhesion of *Navicula* and *Chlorella vulgaris* on the coating surfaces. The quantity of algae was counted in five different regions, and the average density of algae was calculated as a result.

2.8. Actual Marine Test

The Sylgard 184 (used as the reference), Si-PTU, and FSi-PTU coatings were painted on the epoxy resin plates ($100 \times 100 \times 3$ mm). Then, all the samples were mounted on the iron frame and immersed in the South China Sea ($21^{\circ}53' \text{ N}$, $108^{\circ}23' \text{ E}$) (at a depth from 2.0 to 2.5 m). The samples were taken out every month, and photos were taken to record the attachment of marine organisms. Then, Image J software was used to measure the corresponding biofouling coverage five times, and the average biofouling coverage was calculated as a result. The immersion test was conducted between October 2024 and January 2025. An uncoated epoxy resin plate was used as the control.

3. Results and Discussion

3.1. FT-IR Characterization

Figure 2 shows the FT-IR spectra of the FTOH, Si-PTU, and FSi-PTU. The absorption peak at 3312 cm^{-1} is attributed to the -NH- and -OH groups [7,35,36]. The bands at 2935 cm^{-1} , 1731 cm^{-1} , and 1650 cm^{-1} are attributed to the -CH₂-, C=O, -COO-, and -NH- groups [7,35,37,38], respectively. The absorption peak at 1520 cm^{-1} is attributed to the -NHCOS- groups [38]. The bands at 1214 and 1135 cm^{-1} are attributed to the C-F groups [36]. The bands at 1015 cm^{-1} and 800 cm^{-1} are attributed to the Si-O-Si and C-Si groups [32,39]. The infrared spectra confirm that the FSi-PTU polymer has been successfully prepared.

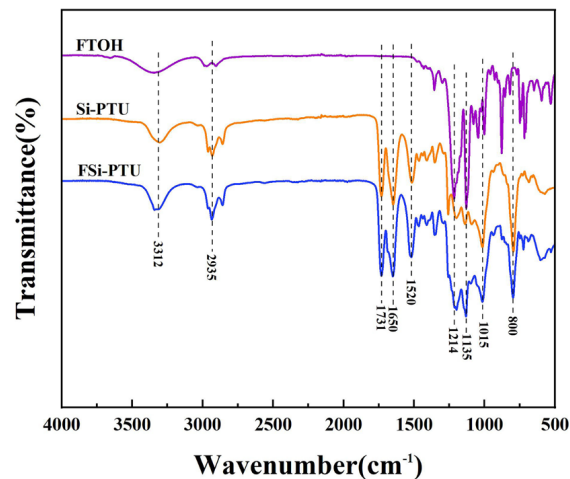


Figure 2. FT-IR spectra of the FTOH, Si-PTU, and FSi-PTU.

3.2. Elemental Characterization

Figure 3 shows the elemental distributions (carbon, oxygen, nitrogen, sulfur, silicon, and fluorine) of the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces. The element signal intensity values of the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces are listed in Table 1. The carbon, oxygen, and silicon elements are evenly distributed on the Sylgard 184 coating surface, as shown in Figure 3(a,a1,a4). The signal intensity of oxygen on the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces is generally the same, as shown in Figure 3(a1–f1). However, the signal intensities of carbon and nitrogen on the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces exhibit clear changes, as shown in Figure 3(a–f,a2–f2). The sulfur signals on the Si-PTU and FSi-PTU coating surfaces gradually reduced with

the increase in the amount of FTOH added, as can be observed in Figure 3(b3–f3). Note that after the introduction of FTOH, clear changes in fluorine signals on the FSi-PTU coating surfaces occurred, as observable in Figure 3(c5–f5). There is an inhomogeneous distribution of the sulfur, silicon, and fluorine signals after the introduction of FTOH, caused by the phase incompatibility between siloxane segments and fluorinated side chains and the hydrophobic and hydrophilic segments of the FSi-PTU polymers, as shown in Figure 3(b3–f3, b4–f4, c5–f5). This reveals that the elemental distribution of the coating surface can be controlled by the amount of FTOH added.

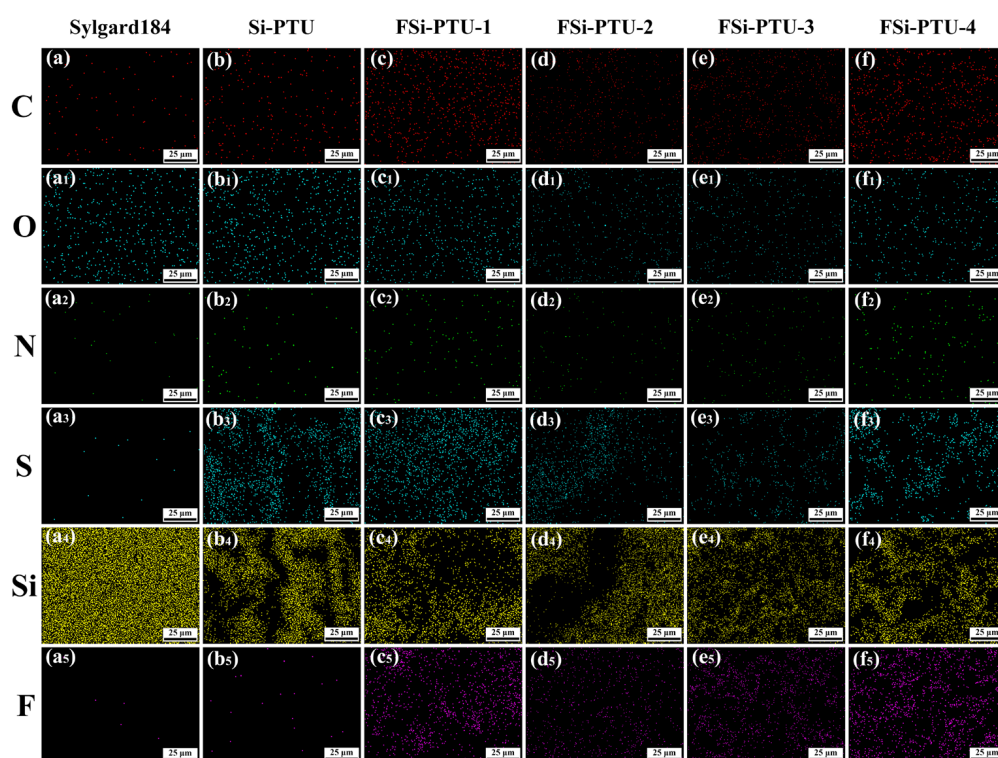


Figure 3. The elemental distribution of carbon (a–f), oxygen (a1–f1), nitrogen (a2–f2), sulfur (a3–f3), silicon (a4–f4), and fluorine (a5–f5) on the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces.

Table 1. The element signal intensity of the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces.

Elt.	Sylgard 184	Si-PTU	FSi-PTU-1	FSi-PTU-2	FSi-PTU-3	FSi-PTU-4
C	68.89	135.90	210.38	186.16	196.74	190.07
O	237.26	194.65	177.38	162.57	147.14	135.28
N	8.15	13.67	17.02	16.09	9.79	13.97
S	0.00	345.64	289.84	243.06	158.62	251.14
Si	2221.81	736.22	510.21	692.30	608.57	426.28
F	1.64	0.10	268.12	229.76	314.65	305.28

3.3. Topography and Surface Roughness

Figure 4 shows the AFM images and Ra values for the coating surfaces. Relatively smooth surfaces are observed for the Sylgard 184 and Si-PTU coatings, with Ra values of about 2.75 and 3.76 nm, as shown in Figures 4a and 4b, respectively. In terms of the results after the introduction of FTOH, a clear increase in the surface roughness of the FSi-PTU coatings can be observed in Figure 4c–f. The Ra values of the FSi-PTU coatings are 33.80, 43.53, 91.33, and 128.67 nm, respectively. The surface roughness of the FSi-PTU coatings increases with the increase in the amount of FTOH added. The coatings exhibit a clear

phase separation phenomenon due to the incorporation of the FSi-PTU coatings' highly polar fluorinated side chains. An appropriate roughness can reduce the contact points between fouling organisms and the coating surface, decrease the adhesion strength, and enhance the antifouling properties. The results show that the surface roughness of the FSi-PTU coatings can be controlled by the amount of FTOH added.

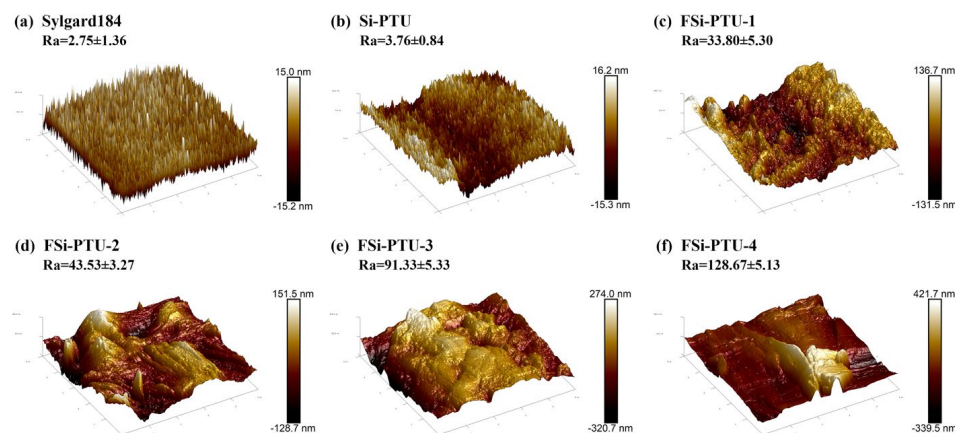


Figure 4. AFM images and Ra values of the Sylgard 184, Si-PTU, and FSi-PTU coating surfaces.

3.4. Water Contact Angle and Surface Energy

Figure 5a shows the water contact angles (WCAs) and diiodomethane contact angles (DCAs) of the Sylgard 184, Si-PTU, and FSi-PTU coatings. The WCAs of the coating surfaces are 113.57, 88.39, 106.05, 106.88, 113.11, and 116.02°, respectively. The DCAs of the coating surfaces are 76.65, 66.39, 65.60, 71.22, 77.35, and 78.55°, respectively. Note that a clear increase in the WCAs and DCAs of the FSi-PTU coatings appears after the introduction of FTOH. Figure 5b shows that the SEs of the coating surfaces are 19.28, 28.82, 25.52, 22.87, 18.94, and 18.25 mN/m, respectively. As the quantity of added FTOH and the surface roughness gradually increase, the SE exhibits a downward trend. This is due to the fact that the hydrophobicity of the coatings is determined by the hydrophobic segments and the surface roughness of the coatings. The hydrophobicity increases with the increase in the amount of the FTOH added. Therefore, the results show that the FSi-PTU coatings exhibit excellent hydrophobic properties after FTOH introduction.

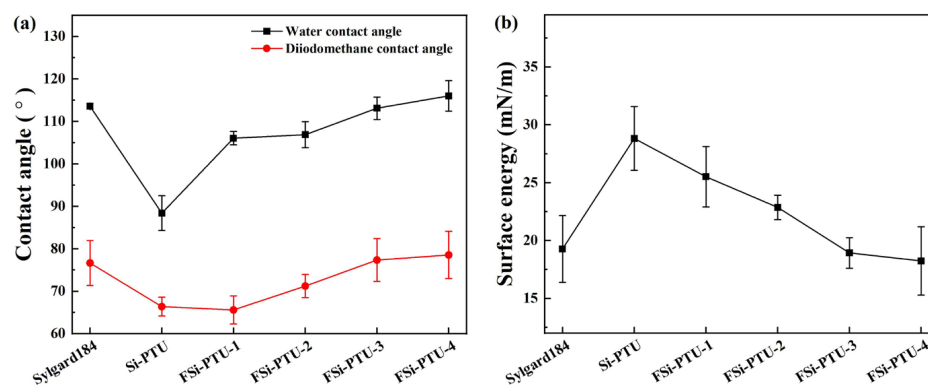


Figure 5. The CA (a) and SE (b) values of the Sylgard 184, Si-PTU, and FSi-PTU coatings.

3.5. Tensile Test

Figure 6 shows the compressive stress–strain curves of the Sylgard 184, Si-PTU, and FSi-PTU polymers. The tensile stress and elongation at break values of the Sylgard 184, Si-PTU, and FSi-PTU polymers are listed in Table 2. The Sylgard 184 polymer exhibits

a lower tensile stress and higher elongation at break, while the Si-PTU polymer shows a relatively higher tensile stress and lower elongation at break. With increasing FTOH addition, the tensile stress of the FSi-PTU polymers decreases and then increases, while the elongation at break increases and then decreases. The C=O groups (-COO-, -NHCOS-, and -NHCOO- groups) in the Si-PTU polymer likely promote the formation of hydrogen bonds, enhance intermolecular interactions, restrict molecular mobility, and increase tensile stress. However, the addition of FTOH introduces fluorinated groups with strong polarity, which attenuates hydrogen-bonding interactions. This results in a reduction in the tensile stress of the FSi-PTU polymers. Conversely, as the FTOH content further increases, the rigidity of the polymers increases, thereby enhancing the mechanical properties of the polymers. Thus, the FSi-PTU polymers exhibit superior mechanical properties compared to conventional silicone-based materials.

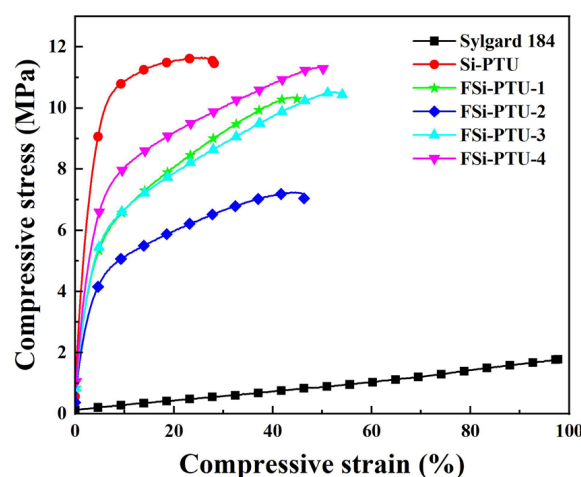


Figure 6. Compressive stress–strain curves for the Sylgard 184, Si-PTU, and FSi-PTU polymers.

Table 2. The tensile stress and elongation at break values of the Sylgard 184, Si-PTU, and FSi-PTU polymers.

Test	Sylgard 184	Si-PTU	FSi-PTU-1	FSi-PTU-2	FSi-PTU-3	FSi-PTU-4
Tensile stress (MPa)	1.74 ± 0.06	10.69 ± 0.41	10.16 ± 0.18	7.19 ± 0.16	10.44 ± 0.33	11.14 ± 0.16
Elongation at break (%)	97.84 ± 3.51	26.88 ± 2.48	44.45 ± 3.68	47.47 ± 3.27	53.64 ± 2.40	50.73 ± 2.32

3.6. Adhesion Test

Figure 7 shows the coatings' adhesion strength values compared to epoxy resin plates. The adhesion strength of the Sylgard 184 coating is only 0.41 MPa, while that of other coatings exceeds 2.14 MPa and meets the requirements for use in the actual marine environment. Hydrogen bonds are a key factor in determining the adhesion strength of the coatings. The FSi-PTU polymer contains a large number of C=O groups, which can form numerous hydrogen bonds with the substrate. As a result, the coating exhibits good adhesion strength to the substrate. However, as shown in Figure 7, the adhesion strength of all FSi-PTU coatings is weaker than that of the Si-PTU coating. The incorporation of fluorinated groups, which effectively lowers the surface tension of the polymer, results in a lower number of C=O groups that can interact with the substrate surface, weakening the adhesion strength of the coatings.

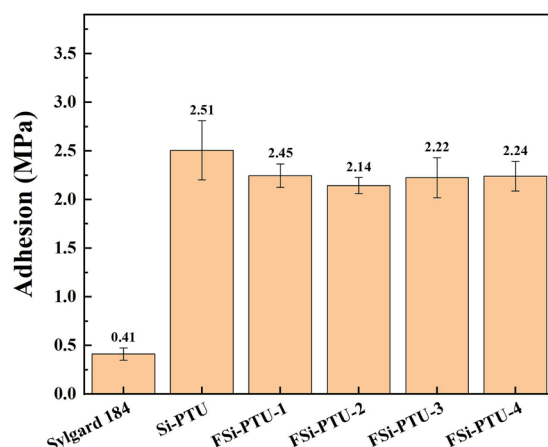


Figure 7. Adhesion strength of the Sylgard 184, Si-PTU, and FSi-PTU coatings.

3.7. Anti-Protein Adsorption

The anti-protein adsorption properties of the coatings were studied and observed using a fluorescence microscope, and the results are shown in Figure 8. A large proportion of green regions can be observed on the Sylgard 184 and Si-PTU coating surfaces, demonstrating the adsorption of a great quantity of BSA on the coating surfaces, as is especially evident in Figure 8a,b. The coatings exhibit relatively poor anti-protein adsorption properties. Conversely, substantially fewer green regions can be seen on all the FSi-PTU coating surfaces. This is mainly due to the effect of the fluorinated side chains, which reduce the adsorption of the coatings to BSA. The FSi-PTU coatings exhibit clear antifouling properties against BSA. Note that Figure 8d,e shows a smaller green area on the FSi-PTU-2 and FSi-PTU-3 coating surfaces, indicating that these coatings have excellent anti-protein adsorption properties against BSA. Overall, the results show that the FSi-PTU coatings have good anti-protein adsorption properties.

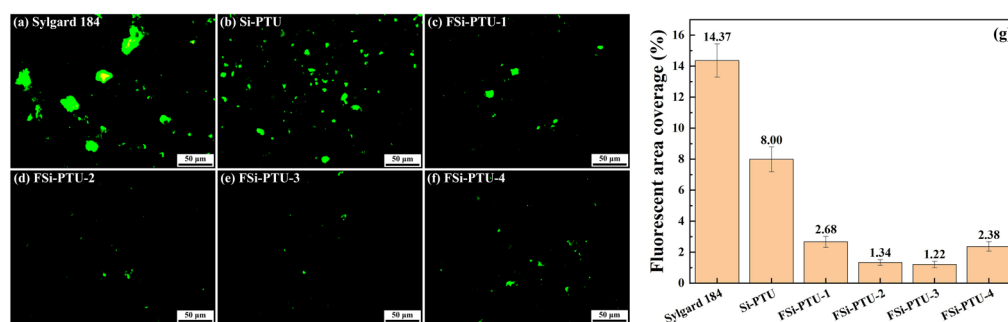


Figure 8. Fluorescence microscopy images (a–f) and the corresponding fluorescent area coverage (g) of the Sylgard 184, Si-PTU, and FSi-PTU coatings after 12 h immersion in BSA-FITC solution.

3.8. Navicula and Chlorella Vulgaris Colonization Assays

Figure 9(a–a₆) shows the optical microscope images and algae density of the coatings after 14 days of immersion in Navicula suspension. It can be observed that there are different amounts of Navicula on the respective coating surfaces after 14 days of immersion. Figure 9(a,a₁) shows that the Sylgard184 and Si-PTU coatings have a poor restraining effect against Navicula. As shown in Figure 9(a₂–a₅), a clear reduction in the number of Navicula can be observed on the surfaces of the FSi-PTU coatings. This indicates that the FSi-PTU coatings have an effective inhibitory effect against Navicula after FTOH introduction. Specifically, few Navicula can be observed on the surface of the FSi-PTU-2 and FSi-PTU-3 coatings, as shown in Figure 9a₄. Interestingly, as shown in Figure 9(b–b₆), the evaluation results regarding the coatings' resistance to Chlorella vulgaris and adhesion exhibit a

similar trend to that for *Navicula*. This is because, after the addition of fluorinated side chains, the SE of the FSi-PTU coatings is reduced, which further weakens the interaction force between the algae (*Navicula* and *Chlorella vulgaris*) and the surfaces of the FSi-PTU coatings. The attenuation of this interaction force promotes the detachment of algae from the surfaces of the FSi-PTU coatings and improves the anti-algal adhesion properties of the FSi-PTU coatings.

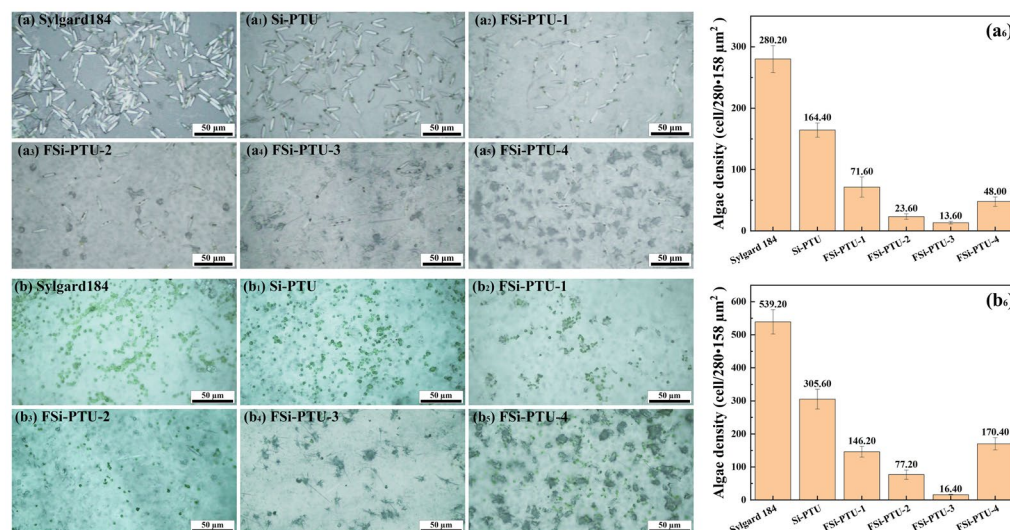


Figure 9. Optical microscope images and algae density of the Sylgard 184, Si-PTU, and FSi-PTU coatings after being immersed in *Navicula* (a–a₆) and *Chlorella vulgaris* (b–b₆) suspension for 14 days.

3.9. Actual Marine Test

Photos of the tested plates that were immersed in natural seawater for three months and their corresponding biofouling coverage are shown in Figure 10. After three months of immersion, the uncoated epoxy resin plate was heavily covered by barnacles, Hydroids, and other species, showing severe biofouling pressure at the experimental location. All remaining plates were colonized by the biofouling organisms to differing extents, indicating that the Sylgard 184, Si-PTU, and FSi-PTU coatings exhibited restricted antifouling performance under static conditions. The antifouling properties of the FSi-PTU coatings were found to be inadequate under actual marine conditions. Notably, with the exception of barnacles, biofouling on the surface of the FSi-PTU-2 coating could be readily detached via gentle rinsing after three months of static immersion. This phenomenon is likely attributed to the optimal distribution of fluorinated side chains, which effectively reduces the interfacial interactions between organisms and the surfaces of the FSi-PTU coatings, enhancing the antifouling properties of the coating under static conditions. However, the surface of the coating is distributed with a large number of -COO-, -NHCOS-, and -NHCOO- groups. These groups can undergo cross-linking reactions with the adhesive proteins secreted by barnacles. As a result, it becomes easier for barnacles to attach to the coating surface, leading to the coating having poor antifouling properties against barnacles. The results demonstrate that the FSi-PTU-2 coating displayed notable antifouling properties following three months of static immersion. However, the formulation of the FSi-PTU coatings requires further optimization to enhance their antifouling properties.

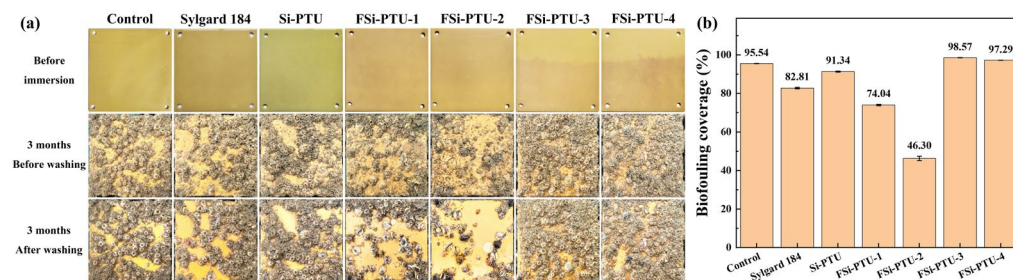


Figure 10. Photos (a) and the corresponding biofouling coverage (b) of plates coated with the Sylgard 184, Si-PTU, and FSi-PTU coatings both prior to and following three months of immersion in the South China Sea.

3.10. Antifouling Mechanism of FSi-PTU

With growing environmental awareness, the application of toxic antifouling coatings has become increasingly restricted, prompting a shift in research efforts toward fouling-release coatings, which are eco-friendly antifouling materials with a low SE [39,40]. In this study, FSi-PTU polymers were synthesized via a polymerization reaction involving HO-PDMS-OH, PETMP, HDI, and FTOH. The polymers exhibit a network structure incorporating siloxane segments and fluorinated side chains. The results show that the FSi-PTU coatings exhibit certain antifouling properties. Figure 11 shows a schematic illustration of the antifouling mechanism of the FSi-PTU coatings. The fluorinated side chains (CF_2 and CF_3 groups) in the FSi-PTU coatings exhibit a heterogeneous distribution on the surfaces. The fluorinated side chains effectively mitigate the hydrogen bonding interaction between marine fouling organisms and the surfaces of the FSi-PTU coatings. Consequently, marine organisms can be readily detached from the surface of the FSi-PTU coatings by the water flow. Barnacles secrete a series of adhesive proteins, which are secreted through ducts to the interface between the barnacles and the substrate. After the replacement of the water layer, these proteins cross-link and solidify with the surface, eventually enabling the barnacles to firmly attach to the underwater interface. The surface of the FSi-PTU coating features a large number of $-\text{COO}-$, $-\text{NHCOS}-$, and $-\text{NHCOO}-$ groups. These groups can react and cross-link with the adhesive proteins secreted by barnacles, thus facilitating barnacle attachment to the FSi-PTU coating surface. This interaction accounts for the FSi-PTU coating's poor antifouling performance against barnacles. In other words, the FSi-PTU coatings can effectively inhibit the attachment of marine biofouling organisms (excluding barnacles) to the coating surface. The approach presented in this study can be readily applied to fabricate the FSi-PTU coatings. Therefore, the antifouling properties of the FSi-PTU coatings may be further improved through the introduction of eco-friendly antifoulants in future studies. This study can also provide a valuable reference for the design of eco-friendly marine antifouling coatings.

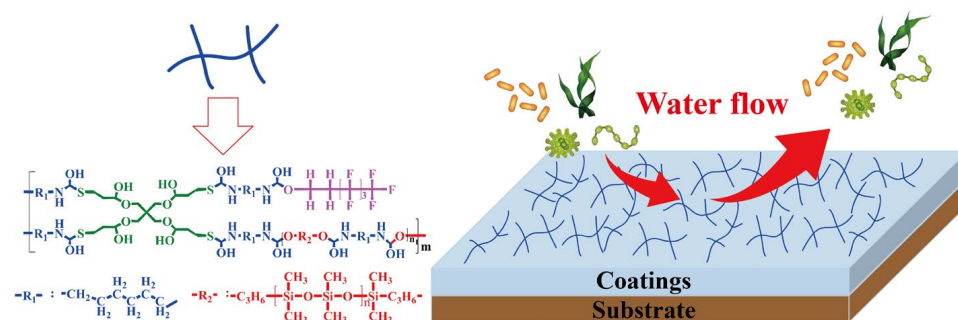


Figure 11. Schematic antifouling mechanism of the FSi-PTU coatings.

4. Conclusions

In this study, an eco-friendly FSi-PTU antifouling coating was prepared through a polymerization reaction with HO-PDMS-OH, FTOH, HDI, and PETMP. This FSi-PTU polymer incorporates siloxane segments and fluorinated side chains, which have an inhomogeneous distribution on the FSi-PTU coating surface, resulting in a hydrophobic surface. The fluorinated side chains effectively mitigate the hydrogen bonding between marine fouling organisms and the FSi-PTU coating surfaces. Marine fouling organisms are readily detached during settlement and adhesion on the surface of the FSi-PTU coatings. The results confirm that these coatings exhibit excellent anti-protein adsorption and anti-algal attachment properties and certain antifouling properties, as observed during an actual three-month-long marine test. The FSi-PTU coatings exhibit a poor antifouling effect against barnacles. Thus, in future research, we plan to incorporate natural antifouling agents (such as capsaicin and betaine) to further enhance the antifouling properties of the coatings. The results provide a reference for the design and development of eco-friendly marine antifouling coatings.

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